

1. Administrative & Study Identification

Full Study Title: A Randomized, Open Label Study to Compare the Effect of Once Monthly Administration of Subcutaneous Mircera Versus Epoetin Alfa on Maintenance of Hemoglobin Levels, Safety and Tolerability in Dialysis Patients With Chronic Renal Anemia
Short Title/Acronym: A Study of Subcutaneous Mircera for the Maintenance Treatment of Dialysis Patients With Chronic Renal Anemia
Protocol Number: ML21208 **NCT Number:** NCT00560404 **Posting ID:** 510110912010584999 **Sponsor/Company Name:** Roche (Hoffmann-La Roche) **Trial Status:** COMPLETED **Investigational Product:** Mircera (Continuous Erythropoietin Receptor Activator / C.E.R.A.) vs. Epoetin Alfa **Phase of Development:** Phase 3 **Medical Monitor:** Clinical Operations Lead, Roche

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To compare the efficacy and safety of once-monthly administration of subcutaneous Mircera versus epoetin alfa for the maintenance of stable hemoglobin (Hb) levels in dialysis patients with chronic renal anemia. **Secondary Objectives:** To evaluate the mean change from baseline in hemoglobin, mean time spent in the hemoglobin target range, and the percentage of participants maintaining their individual hemoglobin concentration throughout the study periods.

2.2 Background & Rationale To address the clinical burden of frequent erythropoiesis-stimulating agent (ESA) dosing in hemodialysis-dependent patients. Patients currently receiving short-acting epoetin alfa require frequent administration. Switching to a continuous erythropoiesis receptor activator (C.E.R.A. / Mircera) allows for a once-monthly dosing regimen, potentially providing stable hemoglobin control with fewer dose changes, thereby simplifying anemia management and improving the standard of care for dialysis patients.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Epoetin alfa) **Blinding:** Open-label
Randomization: Randomized (2 arms)

3.2 Planned Enrollment & Duration Targeted Enrollment: 233 individuals (Target sample size initially estimated at 100 - 500 individuals) **Number of Sites:** Multicenter **Estimated Study Start:**

2007 **Estimated Primary Completion:** 36 weeks post-enrollment
(Anticipated time on study treatment is 3-12 months)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adult patients, ≥ 18 years of age. 2. Chronic renal anemia. 3. Regular hemodialysis with the same schedule of dialysis for ≥ 12 weeks. 4. Maintenance therapy with subcutaneous epoetin alfa at the same administration interval for 4 weeks.

4.2 Exclusion Criteria 1. Transfusion of red blood cells during previous 2 months. 2. Poorly controlled hypertension requiring interruption of epoetin alfa in previous 6 months. 3. Acute or chronic bleeding. 4. Active malignant disease (except non-melanoma skin cancer).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Patients are randomized to receive either monthly injections of Mircera (starting dose 120, 200, or 360 micrograms derived from the previous epoetin alfa dose) or to continue their current epoetin alfa treatment regimen. **Active Comparator:** Epoetin alfa (Trade names: Procrit, Epogen, Eprex, Retacrit). **Route of Administration:** Subcutaneous **Duration of Treatment:** 36 weeks in total, consisting of a Dose Titration Period (Week 0 to Week 28) followed by an Efficacy Evaluation Period (Week 29 to Week 36).

5.2 Concomitant & Prohibited Therapy Permitted: Standard hemodialysis therapies and iron supplementation to support erythropoiesis. **Prohibited:** Concurrent use of any other non-assigned erythropoiesis-stimulating agents (ESAs) during the trial.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Week -4 to -1	Safety Verification Period, baseline Hb concentration assessment.
Baseline	Week 0	Randomization, First Dose Administration, Start of Dose Titration Period.
Interim	Week 1 to 28	Dose Titration Period (DTP): Safety monitoring, Hb level assessment, and permitted dose adjustments.
End of Study	Week 29 to 36	Efficacy Evaluation Period (EEP): Final Hb maintenance measurement, RBC transfusion tracking, IP Accountability.

7. Outcome Measures (Endpoints)

7.1 Primary Outcome Definition: Percentage of Participants Maintaining Their Mean Hemoglobin Concentration Within Plus or Minus 1 Gram/Deciliter of Their Reference Hemoglobin and Between the Target Range During Efficacy Evaluation Period. The target hemoglobin (Hb) range was defined as Hb concentration (gram/deciliter [g/dL]) between 10.5 and 12.5 g/dL during the efficacy evaluation period (EEP). EEP was from Week 29 to Week 36.

Measurement Method: Central/local laboratory analysis of whole blood hemoglobin.

7.2 Secondary Outcomes 1. Mean Change From Baseline in Hemoglobin Concentration Between Baseline and at the Efficacy Evaluation Period. 2. Percentage of Participants Maintaining Individual Hemoglobin Concentration Within the Range of 10.5 - 12.5 Gram/Deciliter Throughout the Efficacy Evaluation Period. 3. Mean Time Spent in Hemoglobin Range of 10.5 - 12.5 Gram/Deciliter During the Efficacy Evaluation Period.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous evaluation of Hb fluctuations, cardiovascular risk factors, and thrombosis events associated with ESA therapy. **Serious Adverse Events (SAEs):** Standard reporting timeline (typically 24 hours). **Data Monitoring Committee (DMC):** Internal/External committee details governing patient safety.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy, Safety Set for adverse events. **Sample Size Justification:** Powered to demonstrate clinical non-inferiority/efficacy of once-monthly Mircera versus standard short-acting epoetin alfa maintenance for the targeted enrollment of 233 participants. **Handling of Missing Data:** Last Observation Carried Forward (LOCF) for continuous laboratory endpoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Routine onsite and remote monitoring visits evaluating adherence to dose titration algorithms. **Audit Trail:** eCRF locking and tracking of dose adjustment justifications.

11. Study Identifiers & Governance

Primary ID: ML21208 **Registry Number:** NCT00560404 **Posting ID:** 510110912010584999 **Sponsor/Lead Organization:** Roche **Study Title:** A Study of Subcutaneous Mircera for the Maintenance Treatment of Dialysis Patients With Chronic Renal Anemia **Official Regulatory Title:** A Randomized, Open Label Study to Compare the Effect of Once Monthly Administration of Subcutaneous Mircera Versus Epoetin Alfa on Maintenance of Hemoglobin Levels, Safety and Tolerability in Dialysis Patients With Chronic Renal Anemia

12. Clinical Framework (Anemia Specific)

Disease Area: Anemia **Primary Condition:** Chronic Renal Anemia **Preferred UMLS Name:** Anemia of chronic kidney disease **Diagnosis Threshold:** ESA-dependent hemoglobin depletion **Medical Methodology:** Erythropoiesis-stimulating agent (ESA) therapy **Optimization Target:** Stable Hemoglobin (Hb) target range of 10.5 - 12.5 g/dL

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for anemia in hemodialysis-dependent CKD **Education Protocol (HCP):** Dose titration algorithm and protocol adherence training **Education Protocol (Patient):** Counseling on anemia maintenance and monthly subcutaneous injection schedule **Quality Audit Mechanism:** Tracking of dose adjustments, Hb tracking compliance, and RBC transfusion events

14. Observation & Follow-Up Schedule

Total Duration: 36 Weeks **Onsite Visit Cadence:** Weeks 0, 4, 8, 12, 16, 20, 24, 28, 32, 36 (Aligning with dialysis/monthly dosing) **Remote Monitoring Frequency:** N/A (Monitored directly via dialysis centers) **Checklist Review Interval:** Every 4 weeks (to assess Hb levels and calculate required dose adjustments)

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				